

MATH 127 Calculus for the Sciences

Lecture 36

December 1, 2025



Today's lecture

Last time

Definition of the definite integral

Properties of definite integrals

Area between curves

This time Section 5.5

Approximating integrals

Area between curves

Last time we computed the area between two curves, $f(x)$ and $g(x)$, by computing:

- The two points where f and g intersect, say $x = a$ and $x = b$.
- Whether $f(x) \geq g(x)$ or $g(x) \geq f(x)$ on $[a, b]$ (say, $f(x) \geq g(x)$ on $[a, b]$).
- The integral

$$\int_a^b (f(x) - g(x)) dx.$$

We saw that:

- this made sense geometrically (especially if both curves f and g lie above the x -axis).
- if we take $g(x) = 0$, this gives back our typical interpretation of the integral as the area under the curve of f .

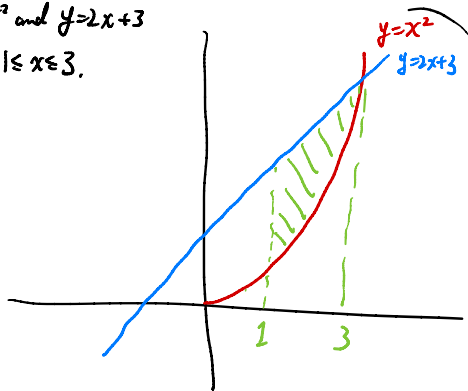
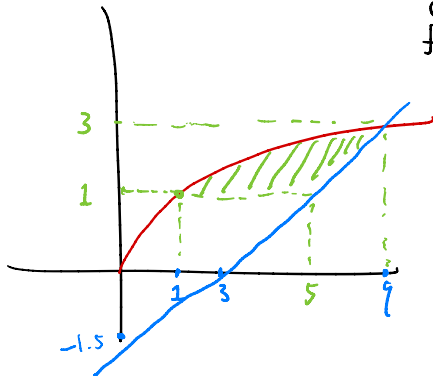
Area between curves: example

Find the area between the curves $y = \sqrt{x}$ and $y = \frac{x-3}{2}$ for $1 \leq y \leq 3$

By flipping the graph, this is the same as the area between

$$y = x^2 \text{ and } y = 2x + 3$$

for $1 \leq x \leq 3$.



$$\int_1^3 (2x+3 - x^2) dx$$

Area under the curve

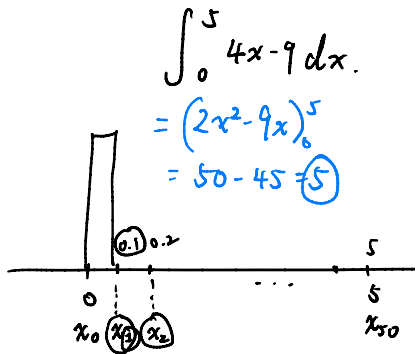
We've seen since the beginning of the course that we can approximate the area under a curve using Riemann sums. Using sigma notation and the following facts:

- $\sum_{i=m}^n cf(i) = c \sum_{i=m}^n f(i).$
- $\sum_{i=m}^n f(i) + g(i) = \sum_{i=m}^n f(i) + \sum_{i=m}^n g(i).$
- $\sum_{i=1}^n 1 = 1 + 1 + 1 + 1 \dots + 1 = n.$
- $\sum_{i=1}^n i = \frac{n(n+1)}{2}.$
- $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$

We can sometimes write the sum just in terms of n . Why is this helpful?

Area under the curve: example

Approximate the area under $f(x) = 4x - 9$ from $x = 0$ to $x = 5$ using $n = 50$ subdivisions. Find two ways to check your estimate.



right Riemann sum = $\sum_{i=1}^{50} \Delta x \cdot f(x_i)$

$$= \sum_{i=1}^{50} 0.1 (4x_i - 9)$$

$$= \sum_{i=1}^{50} 0.1 \left(4 \cdot \frac{i}{10} - 9 \right)$$

$$= \left(0.1 \sum_{i=1}^{50} 4 \cdot \frac{i}{10} \right) - \left(0.1 \sum_{i=1}^{50} 9 \right)$$

$$= \left(0.04 \sum_{i=1}^{50} i \right) - \left(0.9 \cdot \sum_{i=1}^{50} 1 \right)$$

$$= 0.04 \cdot \frac{50 \cdot 51}{2} - 0.9 \cdot 50 = 51 - 45 = 6$$

Error bounds for Riemann sums

For a definite integral $\int_a^b f(x)dx$ and a right Riemann sum R_n (with n subdivisions), which approximates $\int_a^b f(x)dx$, we denote the error in the approximation by

$$E_n = \underbrace{\int_a^b f(x)dx}_{\text{integral}} - R_n$$

Theorem An upper bound for E_n is

$$|E_n| \leq \frac{M}{2n} (b-a)^2$$

where M is the maximum of $|f'(x)|$ on $[a, b]$.

Example

Find an upper bound for $|E_n|$ when approximating $\int_1^2 \frac{1}{x} dx$ with R_1 , R_{10} and R_{100} .

$$\int_1^2 \frac{1}{x} dx = \ln 2$$

$$|E_{100}| \leq \frac{M}{2 \cdot 100} (b-a)^2$$
$$= \frac{M}{2 \cdot 100} (2-1)^2$$

$$= \frac{M}{200} \quad \text{where } M \text{ is max of } \underbrace{|f'(x)|}_{\left|-\frac{1}{x^2}\right|} \text{ over } [1, 2]$$

$$\text{So } M = \max \frac{1}{x^2} \text{ over } [1, 2]$$

Since $\frac{1}{x^2}$ is decreasing, max is obtained at 1.

$$M=1.$$

$$\text{Conclusion } |E_{100}| \leq \frac{1}{200}$$

Example

Find an upper bound for $|E_{10}|$ when approximating $\int_1^2 \frac{1}{x} dx$, $\int_1^e \frac{1}{x} dx$, and $\int_1^4 \frac{1}{x} dx$.